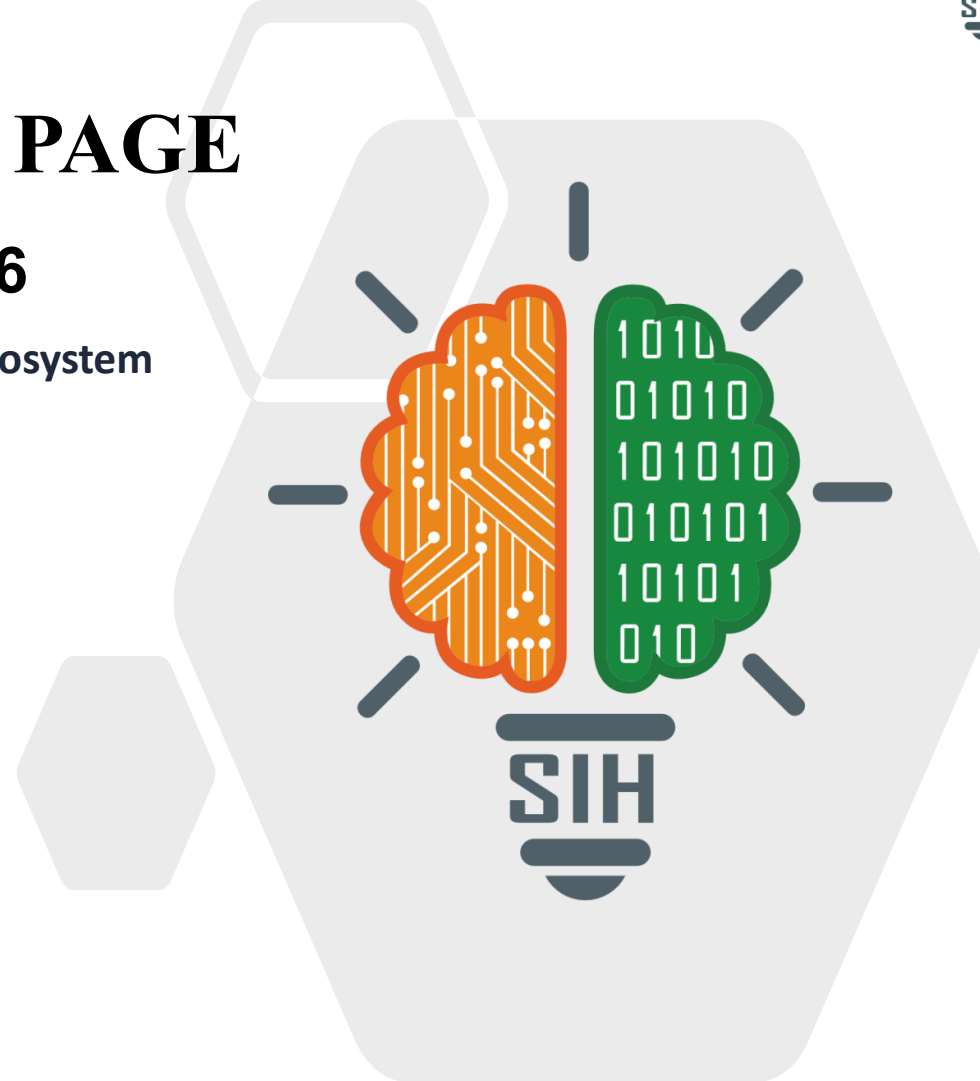


TITLE PAGE

- **Problem Statement ID – SIH26176**
- Problem Statement Title- ORCA: Marine Ecosystem Reasoning with Collaborative Agents.
- **Theme- Miscellaneous**
- **PS Category- Software**
- **Team ID-**
- **Team Name - CodeCatalysts**



DETAILED SOLUTION

- **Operational Ocean Platform:** Real-time agentic system delivering zero-hallucination decision support for coastal safety.
- **Conversational Marine Copilot:** Natural queries via Text & Voice in 12+ Indic languages (Tamil, Malayalam, Hindi, Telugu, etc.).
- **8 Collaborative AI Agents:** Dynamic orchestration decomposing complex queries across domain specialists.
- **Multi-Source Telemetry:** Live satellite (NOAA GHRSSST L4, ISRO OCM-3), weather models (Open-Meteo), and in-situ buoys (INCOIS).
- **Maritime Cockpit HUD:** 60 FPS Windy canvas fluid simulation, turn-by-turn fairway routing, and GMDSS distress SOS beacon.

HOW IT ADDRESSES THE PROBLEM

Bridges fragmented maritime portals into unified, life-saving intelligence:

- **Unifies Fragmented Feeds:** Aggregates scattered data from IMD, INCOIS, ISRO, NOAA & Coast Guard into one pane.
- **Eliminates Manual Search:** Instant, authoritative answers in <1.5s, eliminating multi-portal searching.
- **Craft & Vessel Aware:** Tailors safety guidance specifically for vessel type (artisanal skiffs vs commercial trawlers).
- **Simplifies Complexity:** Converts raw GRIB rasters and buoy metrics into clear Douglas sea-state advisories.
- **Multi-Modal Delivery:** Delivers 60 FPS vector maps, safe fairway corridors, risk scores, and evidence tables.

INNOVATION & UNIQUENESS

- **8 Collaborative AI Agents**
1 Planning + 7 Domain Agents coordinating asynchronously with zero hallucination.
- **Native Indic Voice Copilot**
Full STT/TTS in 12+ coastal languages tailored for low-literacy seafarers.
- **60 FPS Fluid Canvas HUD**
Windy-style real-time particle streams & SST thermal fronts over vector maps.
- **Deterministic Safety & Routing**
0–100 Risk Engine + Cape Comorin fairway corridors (0% land crossing guarantee).
- **Automated GMDSS Distress Dispatch**
Instant Coast Guard MRCC routing (Kochi/Mumbai/Chennai) with SHA-256 receipts.

FACTS & FIGURES — WHY IT MATTERS

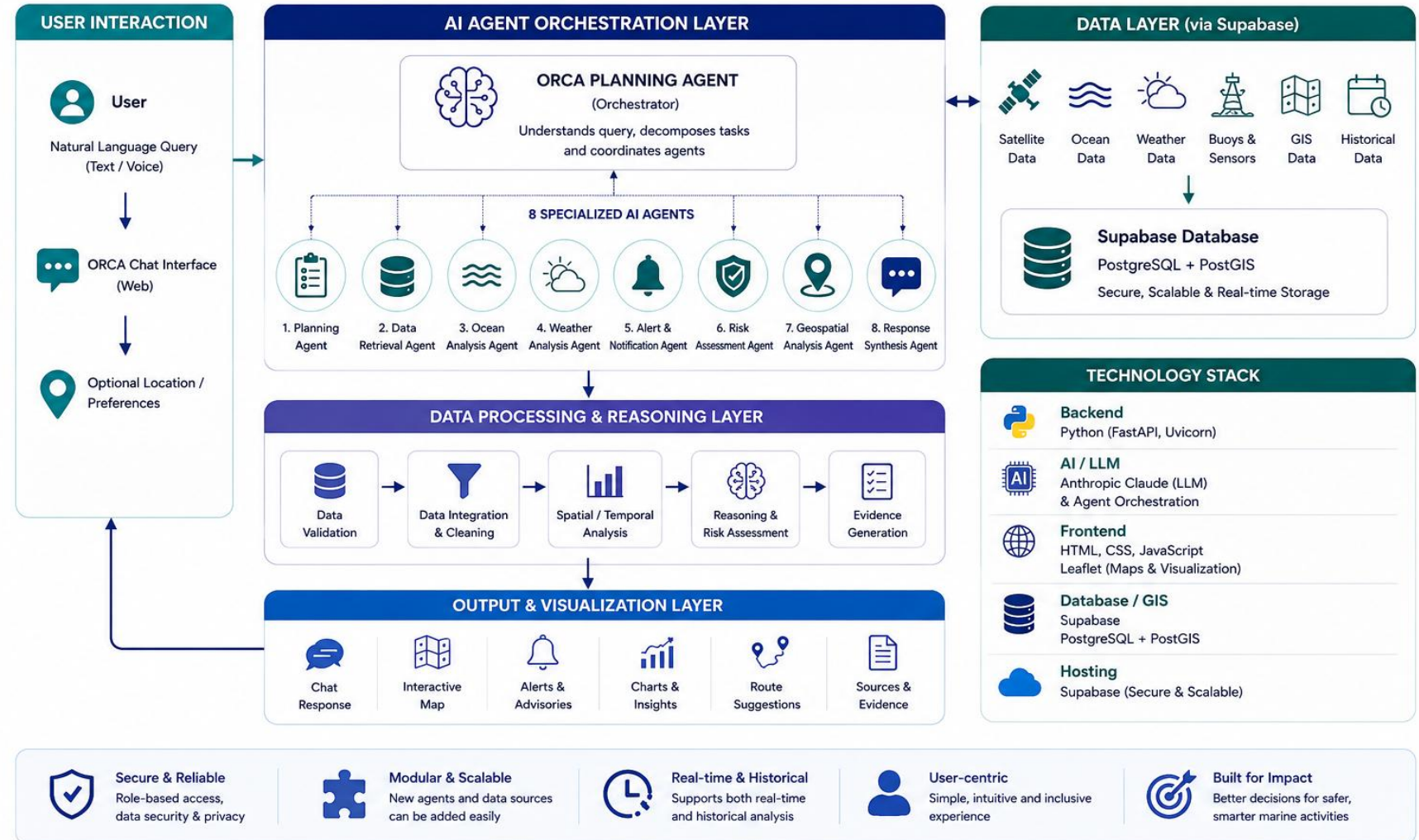
 11,099 km India's revised coastline Source: Survey of India, 2025	 18% Population living in 72 coastal districts Source: Sagarmala Programme	 ~80 million People depend on the sea for their livelihoods Source: Govt. of India, 2026	 28.06 million+ Fishermen in India Source: Ministry of Fisheries, 2024	 95% (by volume) 70% (by value) India's trade moved through maritime routes Source: MoPSW, 2024	 ₹62,408 crore Seafood export earnings (FY 2024–25) Source: MPEDA, 2024–25
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WHO NEEDS MARINE INTELLIGENCE?

 Fishing Communities Fishing zones, safety, planning and advisories	 Researchers & Academia Ocean analysis, trends, data for research	 Government & Authorities Hazards, alerts, monitoring & enforcement	 Maritime Operators Routes, weather, sea conditions & port operations	 Environmental Stakeholders Sensitive areas, conservation & sustainability
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IMPLEMENTATION FLOW

- 1 USER INPUT**
Natural text/voice + auto GPS location
- 2 PLANNING**
Intent classification & agent delegation
- 3 8 AI AGENTS**
Specialized domain multi-agent team
- 4 DATA RETRIEVAL**
Satellite + Weather + Buoys + GIS
- 5 VALIDATE & ANALYSE**
Quality control & <12h freshness check
- 6 REASON & ASSESS**
Risk Engine (0–100) + Fairway Routing
- 7 OUTPUT**
60 FPS Windy map + GMDSS + Evidence



FEASIBILITY

- **Reliable Data Sources** — Open-Meteo, NOAA CoastWatch GHRSSST L4, ISRO MOSDAC OCM-3, INCOIS Buoys, IMD.
- **Proven Technologies** — Python 3.13 FastAPI, Supabase PostGIS, Leaflet.js, React 19, PWA offline caching.
- **Modular Multi-Agent Design** — 8 independent domain agents with plug-and-play connector abstractions.
- **Scalable Infrastructure** — Supabase cloud backend + embedded zero-dependency spatial index (<5ms latency).
- **100% Tested MVP** — Fully validated production codebase with 34/34 passing automated tests.

VIABILITY

- **Multi-Stakeholder Value** — Artisanal canoes, mechanized trawlers, merchant vessels, port masters & Coast Guard.
- **Continuous Dynamic Feeds** — Satellite passes refreshed daily (GHRSSST L4) and weather hourly (Open-Meteo).
- **Extensible Maritime Horizons** — Plug-in readiness for Blue Carbon monitoring, mariculture & offshore wind.
- **Rapid National Scalability** — Pilot tested in Kochi, Mumbai, Chennai; ready for all 72 coastal districts.

CHALLENGES & RISKS

HOW WE HANDLE THEM

1. **Diverse Formats & Sources:** → Unified Pydantic v2 schemas & ingestion pipeline
2. **Data Freshness & Lags:** → Freshness penalty (+8 risk if telemetry age > 12h)
3. **Incomplete / Conflicted Data:** → In-situ OMNI buoys cross-validate satellite grid
4. **LLM Hallucinations:** → Pure deterministic math engine (LLM for intent only)
5. **Heavy Spatial Queries:** → Embedded Ray-Casting & Haversine index (<5ms)
6. **Safe Nautical Routing:** → IMO Douglas/Beaufort rules & Cape Comorin fairway
7. **API Downtime / Offline:** → PWA Offline Service Worker & pre-seeded local cache
8. **Privacy & Distress Security:** → Stateless tokens & encrypted GMDSS SHA-256 dispatch

WHO IS IMPACTED?

**FISHING COMMUNITIES**

- Get advisories on potential fishing zones
- Plan safer and more productive trips
- Reduce risk and save time and fuel

**RESEARCHERS &
ACADEMIA**

- Easy access to multi-source marine data
- Accelerates research and insights
- Supports innovation in ocean science

**GOVERNMENT &
AUTHORITIES**

- Real-time alerts for hazards & marine events
- Better monitoring of coastal and maritime activities
- Supports informed policy and rapid response

**MARITIME OPERATORS**

- Plan routes with better weather & sea insights
- Reduce operational risk and delays
- Improve efficiency and resource planning

**ENVIRONMENTAL
STAKEHOLDERS**

- Monitor sensitive marine ecosystems
- Promote sustainable use of ocean resources
- Support conservation and protection efforts

WIDER BENEFITS

**SOCIAL**

- Safer and easier access to marine information
- Better safety awareness
- Stronger coastal communities

**ECONOMIC**

- Better planning and decisions
- Lower avoidable effort and costs
- Supports marine operations

**ENVIRONMENTAL**

- Better awareness of sensitive areas
- Supports sustainable marine activities
- Supports conservation and protection

SCALABILITY

**REGIONAL →
NATIONAL**

Start with a focused coastal pilot and expand to more regions and national scale.

**MODULAR
ARCHITECTURE**

Add new AI agents, datasets and features without rebuilding the core system.

**CLOUD-READY**

Supabase-backed infrastructure is designed to scale with growing users and data.

**MORE DATA →
MORE VALUE**

Adding reliable data sources increases coverage and enables richer insights and use-cases.

RESEARCH BASIS

- Marine Earth Observation (SST Fronts & Radiometry)
- Operational Oceanography (Douglas & Beaufort Scales)
- Geospatial & Nautical Routing (Geodesic Fairways)
- Multi-Agent Systems (Collaborative Delegation)
- Conversational Marine UX (12+ Indic Voice Dialects)
- Evidence & Data Provenance (Audit Latency Ledgers)

DATA SOURCE GROUPS

Satellite Observation

NOAA GHRSSST L4 OSTIA, ISRO Oceansat-3 OCM, Copernicus

Physical Oceanography

SST, Chlorophyll-a, Wave Swell, Ocean Currents & Tides

Meteorological Feeds

Open-Meteo GFS/ECMWF, Wind Vectors, Gusts & Squalls

Safety & Warning Advisories

IMD Marine Warnings, Port Signals 1–11, Damini Lightning

Maritime GIS Boundaries

EEZ 200 NM, IMBL Borders, Marine Protected Areas (MPAs)

Historical & In-Situ Baselines

INCOIS Deep-Sea OMNI Buoys, Coastal Tide Gauges & CMFRI

REFERENCE AREAS

- INCOIS (MoES) — PFZ Advisories & Ocean State Forecasts
- ISRO MOSDAC — Oceansat-3 & INSAT-3D Satellite Data
- IMD (MoES) — Cyclone Warnings & Port Danger Signals 1–11
- Indian Coast Guard — MRCC Search & Rescue Protocols
- NOAA CoastWatch — Global GHRSSST Level-4 Telemetry
- Survey of India & NIOT — Deep-Sea Buoys & Tide Gauges

Production release integrates verified live APIs from Open-Meteo, NOAA CoastWatch, ISRO MOSDAC, and INCOIS.